

DC

Short for “direct current,” an electric current that flows in one direction continuously.

deceleration

A decrease in velocity that occurs when a force pushes against a moving object in the opposite direction to its direction of motion.

decomposition

To break up into two or more simpler ingredients.

deformation

To be changed in shape by a force, such as being stretched, bent, or squeezed.

density

A quantity of how much matter is held within a known volume of a material.

diffraction

A behavior of waves, in which a wave spreads out in a number of directions after it passes through a small gap, with a width similar to its wavelength.

dipole

A molecule with two poles: one negative and one positive.

displacement

The moving aside of part of a medium by an object placed in that medium. Or the distance between one point and another.

distillation

A process that separates liquid mixtures by boiling away each component in turn, then cooling them back into pure liquids.

DNA

Short for “deoxyribonucleic acid,” a complex chained molecule that carries genetic code, the instructions that a cell—and entire body—uses to make copies of itself.

drag

The resistance force formed when an object pushes through a fluid, such as air or water.

dynamic equilibrium

When a reversible reaction takes place at the same rate in both directions so, even though it is continuing in both directions, the overall quantities of the materials involved stay constant.

eclipse

An eclipse occurs when the Earth, Sun, and Moon line up, blocking out the view of one of the objects. In a solar eclipse, the Moon covers up the Sun as seen from Earth. In a lunar eclipse, the Earth sits between the Sun and the Moon.

ecosystem

A collection of living organisms that share a habitat and are reliant on each other for survival.

elasticity

The property of an object that allows it to change shape when forced to but return to its original form when the force is removed.

electrolysis

Dividing compounds into simpler substances using the energy in electricity.

electrolyte

A liquid that conducts electricity.

electromagnet

A magnet that can be turned on by running an electric current through it.

electron

A negatively charged particle that is located around the outside of an atom.

electronics

A field of science and technology that involves using semiconductors to make components for circuits.

element

A natural substance that cannot be divided or simplified into raw ingredients. There are around 90 natural elements on Earth.

endothermy

The ability of an animal to maintain a constant body temperature using energy burned from its food to heat or cool the body.

energy

Energy is what allows things to happen. For example, chemical energy in food allows us to live and move.

enzyme

A protein that is used to control a chemical reaction or other process taking place inside a living body.

evaporate

To turn from a liquid to a gas, such as a puddle drying out.

evolve

A change in the characteristics of a species due to its environment; evolution is driven by a process called natural selection.

exoskeleton

Hard tissue that forms the outer surface of a body, giving shape and structure to it.

exothermic

Describing an animal that does not maintain a constant body temperature but allows it to fluctuate with that of the surroundings.

fat

A solid lipid—a biological material that is used to store energy, insulate nerves, and form membranes. Liquid lipids are called oils.

filtration

The process of passing a substance through a filter to remove solid particles.

fission

Breaking apart; nuclear fission involves radioactive atoms splitting in two, releasing a huge amount of energy.

force

The means that causes a mass to change its momentum.

fossil fuel

A substance that burns easily, releasing heat formed from the remains of ancient plants and other organisms; fossil fuels include coal, natural gas, and oil.

friction

A force that occurs between moving objects, where the surfaces rub against each other, opposing their movement.

fusion

Joining together; nuclear fusion involves two small atoms fusing into a single larger one, releasing huge amounts of energy.

gene

A coded instruction for making a certain body feature that is passed from parent to offspring; the code is stored as a DNA molecule and is translated into proteins, each of which performs a specific job.

generator

A device for converting rotational motion into electric current.

gills

A breathing organ that takes oxygen from the water and releases carbon dioxide. Gills are used by fish and many underwater creatures.

gland

An organ in the body that secretes chemicals in large quantities; endocrine glands release chemicals into the blood stream, exocrine glands secrete onto the surface of the body.